

PHASE

PROJECT DESIGN PHASE

Date	28 September 2026
Team ID	SWTID-2026-5127
Project name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum marks	4

3. Project Design Phase

3.1 Purpose of this phase

The design phase decides *how* the requirements will be met on ServiceNow. It defines the catalog item structure, the flow logic, data usage, notifications, security and user experience, so that development becomes a matter of building what has been designed.

3.2 Solution architecture

Layer	Component	Role
Presentation	Service Portal / Catalog	Where the user finds and submits the request
Request	Catalog item and variables	Captures structured data
Record	Request (REQ), Requested Item (RITM)	Stores the request and its state
Automation	Flow Designer flow and subflows	Orchestrates approval, tasks, updates, notifications
Work	Catalog Task / Task records	Procurement and fulfilment work
Communication	Notifications (email)	Keeps everyone informed
Insight	Reports and dashboards	Measures performance

3.3 Catalog item design

Name: Standard Laptop Request **Category:** Hardware **Short description:** Order a company-standard laptop for yourself or a team member.

Variable	Type	Mandatory	Notes
Requested for	Reference (User)	Yes	Defaults to logged-in user
Laptop model	Select box / Reference	Yes	Only standard models
Justification	Multi-line text	Yes	Why the laptop is needed
Delivery location	Reference (Location)	Yes	Where to deliver
Needed-by date	Date	Yes	Cannot be in the past (client script)
Cost center	String / Reference	Yes	For budgeting
Additional accessories	Checkbox / Select	No	Bag, mouse, docking station

Catalog client script (design intent): validate that the needed-by date is not earlier than today and auto-populate the cost center from the selected user.

Access control: available to all employees through user criteria; managers may order on behalf of team members.

3.4 Flow design

Flow name: Standard Laptop Order Automation **Trigger:** Service Catalog — runs when the catalog item is ordered.

Step	Action	Purpose
1	Trigger: Service Catalog	Starts the flow with the RITM and variables
2	Look Up Record: User (requested for)	Retrieve the manager
3	If: manager is empty	Choose fallback approver

Step	Action	Purpose
4	Update RITM: state = Awaiting Approval	Keep status accurate
5	Ask for Approval: manager	Obtain decision, with rule "approve/reject"
6	If: approval = Approved	Branch on outcome
7	Create Catalog Task: Procurement	Assign to Procurement group with model, cost center and delivery details
8	Send Notification: requester and approver	Inform of approval
9	Wait for Condition: procurement task closed	Pause until ordering is complete
10	Create Catalog Task: IT Setup	Imaging, configuration, handover
11	Wait for Condition: IT task closed	Pause until laptop is delivered
12	Update RITM: state = Closed Complete	Finish
13	Send Notification: completion	Inform requester
Else	Update RITM: Closed Incomplete, add rejection comment, notify requester	Handle rejection

3.5 Flow diagram

```

[Start: Catalog request submitted]
|
[Look up requester's manager]
|
<Manager found?> --No--> [Use fallback approver]
|Yes                      |
+-----+
|
[Set RITM = Awaiting Approval]
|
[Ask for Approval]
|           |
Approved   Rejected
|           |
[Create Procurement Task] [Set RITM = Closed Incomplete]
|           |
[Wait: task closed]       [End]
|
[Create IT Setup Task]
|
[Wait: task closed]
|
[Set RITM = Closed Complete]
|
[Notify requester] --> [End]

```

3.6 Approval design

- **Approver:** manager of the requested-for user.
- **Fallback:** a designated approver group if manager is missing or inactive.
- **Rule:** any single approval decision is sufficient (manager only).
- **Reminder:** a Wait action combined with an "if still pending" check sends a reminder after a set duration, and can escalate after a longer period.
- **Rejection:** comments required; the reason is copied to the RITM and included in the notification.

3.7 State model of the RITM

State	Meaning
Open / Pending	Request submitted
Awaiting Approval	Waiting for manager decision
Work in Progress	Procurement or setup underway
Closed Complete	Laptop delivered
Closed Incomplete	Request rejected or cancelled

3.8 Notification design

Event	Recipient	Content
Request submitted	Requester	Confirmation with request number
Approval needed	Manager	Request summary with approve/reject link
Reminder	Manager	Pending approval reminder
Approved	Requester	Approval confirmation and next steps
Rejected	Requester	Rejection with reason
Task assigned	Procurement / IT group	Task details
Completed	Requester	Laptop ready or delivered

3.9 Reusability: subflows and actions

To keep the main flow readable and to support future requests (monitor, phone, software), the design proposes:

- **Subflow "Manager Approval with Fallback"**: looks up the manager, applies fallback logic and returns the decision.
- **Subflow "Create Fulfilment Task"**: accepts group, description and due date.
- Flow variables and data pills are used instead of hard-coded values wherever possible.

3.10 Security and access design

- Catalog item is visible to employees but editable only by catalog administrators.
- Flow editing is limited to users with flow designer roles.
- Approvers can act only on records assigned to them.
- Procurement and IT groups see only their tasks.
- Personal data in the request is limited to what is needed.

3.11 Error handling design

- Flow error handling notifies the administrator group when an action fails.
- Execution details are reviewed in the Flow Designer execution log.
- Missing data (for example, no manager and no fallback) sets the RITM to a "needs attention" state and assigns it to the helpdesk.

3.12 User experience design

- Short, clear form with helpful labels and tooltips.
- Default values reduce typing.

- The portal shows request status, approval outcome and task progress in one place.
- Notification wording is simple and action-oriented.

3.13 Reporting design

A dashboard with: number of requests per month, average approval time, average fulfilment time, rejection rate and requests by laptop model. These give management evidence of improvement.

3.14 Design decisions and rationale

Decision	Rationale
Use Flow Designer instead of scripts	Low-code, visual, easier to maintain and audit
Restrict to standard models	Simplifies procurement and support
Manager-only approval	Matches current policy; keeps the flow simple
Use subflows	Enables reuse for other request types
Notifications at each milestone	Reduces "status" inquiries

3.15 Outcome of the phase

A complete design covering the catalog item, flow logic, approval and notification behaviour, state model, security and reporting. Development can now be planned and executed against this blueprint.